AI-POWERED SUPPLY CHAIN DISRUPTION ASSISTANT

Dataset Details

Supply Chain Disruption Recovery Dataset Documentation

File: backend/data/supply_chain_disruption_recovery.csv

Records: 100,000 | Columns: 15 | Missing values: 0

Duplicates: 0 | Profile: evaluation/dataset_profile.json

1. Dataset Overview

The dataset is backend/data/supply_chain_disruption_recovery.csv, used as the source for MySQL ingestion, ChromaDB + BM25 index construction, offline model training, and historical analytics. No external publisher, citation, dataset URL, or explicit data-generation methodology is documented in this repository. The repository does not state whether records are synthetic, operational, or otherwise derived. All figures in this document are sourced from evaluation/dataset_profile.json generated by scripts/inspect_dataset.py.

2. File Location and Size

Property	Value
Relative path	backend/data/supply_chain_disruption_recovery.csv
File size	Approximately 11.5 MB
Vector index location	backend/data/chroma/ (generated by build_vector_index.py)
BM25 index location	backend/rag/bm25_index.joblib (generated by build_bm25_index.py)
Profile artifact	evaluation/dataset_profile.json

3. Row and Column Counts

Property	Value
Total rows	100,000
Total columns	15
Missing values (all columns)	0
Exact duplicate rows	0
Duplicate disruption_id values	0
Valid rows (schema-conforming)	100,000
Profile generated by	scripts/inspect_dataset.py

4. Column Descriptions

Column	CSV Type	App Type	Description
disruption_id	object	string	Source case identifier; used as document ID in ChromaDB.
disruption_type	object	string (enum)	Port Congestion, Geopolitical, Labor Strike, Natural Disaster, Cyber Attack, Factory Incident.
industry	object	string (enum)	Electronics, Automotive, Consumer Goods, Pharmaceuticals, Aerospace.
supplier_tier	int64	string '1'–'4'	Tier 1 = most critical. Stored and used as a string.
supplier_region	object	string (enum)	Asia-Pacific, Europe, North America, South America, Africa/Middle East.
supplier_size	object	string (enum)	Small, Medium, Large.
has_backup_supplier	bool	boolean	Whether a backup supplier was available at event time.
disruption_severity	int64	integer 1–5	Severity: 1 = minor, 5 = critical.
production_impact_pct	float64	float 0–100	Production capacity impact as a percentage.
revenue_loss_usd	float64	float >= 0	Revenue loss in USD.
response_type	object	string (enum)	Alternative Supplier, Production Reroute, Inventory Buffer, Customer Delay, Combined Strategy.
response_time_days	int64	integer >= 0	Days from event detection to active response.
partial_recovery_days	int64	integer >= 0	Days to partial operational recovery.
full_recovery_days	int64	integer >= 0	Days to full operational recovery.
permanent_supplier_change	bool	boolean	Whether a permanent supplier change resulted from the event.

5. Categorical Value Distributions

disruption_type

Value	Count	Share (%)
Port Congestion	25,118	25.1
Geopolitical	18,084	18.1
Labor Strike	15,017	15.0
Natural Disaster	15,016	15.0
Cyber Attack	14,887	14.9
Factory Incident	11,878	11.9

industry

Value	Count	Share (%)
Electronics	29,844	29.8
Automotive	24,934	24.9
Consumer Goods	20,158	20.2
Pharmaceuticals	15,015	15.0
Aerospace	10,049	10.0

supplier_region

Value	Count	Share (%)
Asia-Pacific	40,262	40.3
Europe	21,890	21.9
North America	19,939	19.9
South America	9,896	9.9
Africa/Middle East	8,013	8.0

Other categorical columns

Column	Values (Count)
supplier_tier	Tier 1 (34,938), Tier 2 (30,023), Tier 3 (25,019), Tier 4 (10,020)
supplier_size	Medium (45,161), Small (29,872), Large (24,967)
has_backup_supplier	True (55,574), False (44,426)
response_type	Alternative Supplier (44,362), Production Reroute (18,858), Inventory Buffer (12,687), Customer Delay (12,429), Combined Strategy (11,664)
permanent_supplier_change	False (76,080), True (23,920)

6. Numerical Column Ranges

Column	Min	Max	Mean	Median	IQR Outliers
disruption_severity	1	5	2.50	2.0	3,791
production_impact_pct	1.0%	100.0%	31.41%	30.1%	667
revenue_loss_usd	$10,000	$373,130,977	$2,522,812	$875,466	10,620
response_time_days	1 day	64 days	4.07	2.0	7,094
partial_recovery_days	3 days	337 days	27.51	20.0	5,190
full_recovery_days	8 days	798 days	75.84	61.0	4,090

7. Target Variables

Target Column	Prediction Task	Model Type
production_impact_pct	Estimated production capacity impact (%)	Random Forest Regressor
revenue_loss_usd	Estimated financial revenue loss (USD)	Random Forest Regressor
full_recovery_days	Estimated time to full recovery (days)	Random Forest Regressor
permanent_supplier_change	Risk of permanent supplier change	Random Forest Classifier

8. Input Features

Seven initial-event features used by all four prediction models (columns available before event outcome is known):

Feature	Encoding	Notes
disruption_type	One-hot	6 categories.
industry	One-hot	5 categories.
supplier_tier	Ordinal integer 1–4	Numeric passthrough.
supplier_region	One-hot	5 categories.
supplier_size	One-hot	Small, Medium, Large.
has_backup_supplier	Boolean (0/1)	Direct mitigation capacity indicator.
disruption_severity	Ordinal integer 1–5	Strongest predictor of impact and recovery.

Excluded (label leakage): production_impact_pct, revenue_loss_usd, full_recovery_days, permanent_supplier_change (prediction targets); response_type, response_time_days, partial_recovery_days (post-event outcomes unavailable at prediction time).

9. Data Quality

- Zero missing values across all 15 columns (verified by dataset_profile.json).
- Zero exact duplicate rows.
- Zero duplicate disruption_id values.
- IQR-based outliers are reported in dataset_profile.json but are not automatically removed.
- Ingestion enforces hard bounds: severity 1–5, impact 0–100%, revenue_loss_usd >= 0, duration columns >= 0.

10. Data Preprocessing

During Ingestion (scripts/ingest_dataset.py)

- Schema validation — requires exactly 15 expected columns.
- Text normalization — strips whitespace; normalizes boolean representations (true/false/1/0).
- Boundary enforcement — rejects rows violating hard constraints.
- Deduplication — removes exact duplicates; resolves duplicate disruption_id by keeping first occurrence.
- Batch insertion — SQLAlchemy bulk operations; skips disruption_ids already in MySQL.

During Training (scripts/train_models.py)

- One-hot encoding: disruption_type, industry, supplier_region, supplier_size.
- Numeric passthrough: supplier_tier, disruption_severity.
- Boolean encoding (0/1): has_backup_supplier.
- No imputation — dataset has no missing values.
- No outlier removal — IQR outliers are retained in training data.

11. Training and Test Split

Split	Rows	Percentage	Purpose
Training	70,000	70%	Model parameter fitting.
Validation	15,000	15%	Model selection (Random Forest vs Extra Trees) by MAE or F1.
Test (held-out)	15,000	15%	Final metric measurement; not seen during training or validation.

12. Dataset Usage per ML Task

Task	Target	Features Excluded Beyond Non-Event Columns
Production Impact	production_impact_pct	All other targets and post-response fields.
Financial Loss	revenue_loss_usd	All other targets and post-response fields.
Recovery Time	full_recovery_days	All other targets and post-response fields.
Supplier Change	permanent_supplier_change	All other targets and post-response fields.

13. Retrieval and Indexing Preparation

Each record is rendered into a single textual document containing all 15 attributes in a natural-language format. This document is:

- Embedded by BAAI/bge-small-en-v1.5 (384 dims, L2-normalized) and stored in ChromaDB with categorical metadata for filtered retrieval.
- Tokenized and indexed in a BM25 keyword index persisted at backend/rag/bm25_index.joblib.

One document per case; no chunking or overlap applied.

14. Limitations and Data Provenance

- External source is undocumented — the repository does not record whether the data is synthetic, operational, or publicly sourced.
- No temporal timestamps — prevents temporal trend analysis.
- No supplier identifiers — supplier-entity-level analysis is not supported.
- Outcome correlations are not causal — a historical response does not prove it caused the recovery outcome.
- Revenue loss and recovery time models have moderate accuracy (R2 ~0.37); unsuitable for exact financial planning.